

MID SEM PROJECT REPORT(CSPC 401 & ITPC 401)

MEDBOARD: MULTI-AGENT BRAIN MRI ANALYSIS AND PRELIMINARY REPORTING SYSTEM

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Certificate

This is to certify that the project report entitled
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The work presented in this report is genuine to the best of my knowledge and has not been submitted elsewhere for any other academic purpose.

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Abstract:

Magnetic Resonance Imaging (MRI) is the clinical gold standard for the diagnosis, preoperative evaluation, and longitudinal monitoring of intracranial neoplasms. However, manual radiological interpretation remains labor-intensive, subject to inter-observer variability, and vulnerable to diagnostic bottlenecks caused by a critical shortage of specialized neuroradiologists. Standard computer-aided diagnosis (CAD) systems typically treat tumor classification and lesion segmentation as isolated, single-task problems, lacking clinical interpretability, quantitative morphological extraction, and structured reporting workflows.

To address these limitations, this project presents MedBoard, an integrated, multi-agent artificial intelligence system engineered to process raw 2D axial brain MRI slices through a cohesive diagnostic pipeline. MedBoard unifies automated pre-processing, dual-tier deep perception (high-accuracy 4-class tumor classification and sub-millimeter semantic segmentation), automated morphological biomarker extraction, and an interactive clinician decision-support interface.

Experimental evaluations on the benchmark BRISC 2025 dataset (6,000 axial slices across Glioma, Meningioma, Pituitary Tumor, and Normal scans) demonstrate the efficacy of our proposed architecture. For classification, a transfer-learned EfficientNet-B0 backbone achieves 99.00% test accuracy (Macro-F1: 0.9899, Test Loss: 0.0529) on 1,000 unseen test slices with an inference latency under 15 milliseconds. For semantic segmentation, an extensive benchmark across 860 unseen test slices confirms that a Pretrained ResNet-34 U-Net++ architecture with dense nested skip pathways establishes the top performance, achieving an 87.78% Mean Dice score and 81.11% Mean IoU, outperforming standard baseline U-Net (86.84%), Attention U-Net (86.96%), SaberNet (86.66%), and GeoSample U-Net (86.46%).

Building upon this validated dual-tier perception foundation, the project is actively integrating an automated morphological feature extraction module, a local vector database (ChromaDB) grounded in WHO CNS5 clinical guidelines via Retrieval-Augmented Generation (RAG), and a Large Language Model (Gemini 2.0 Flash) synthesis agent to automatically generate structured preliminary radiology reports. This research establishes a reproducible, defendable foundation for scalable, transparent computer-aided neuro-oncology diagnostics.

Introduction

Brain tumors represent one of the most debilitating categories of human neoplasms, with over 300,000 primary central nervous system (CNS) tumors diagnosed globally each year. Magnetic Resonance Imaging (MRI)—specifically T1-weighted, T2-weighted, and Fluid-Attenuated Inversion Recovery (FLAIR) modalities—serves as the primary non-invasive imaging technique for intracranial tumor localization, characterization, and stereotactic surgical guidance. The key properties that define modern computer-assisted brain MRI analysis systems are:

- **Decoupled Multi-Task Perception:** Accurately categorizing the pathological class of neoplasm (intra-axial glioma, extra-axial meningioma, sellar pituitary adenoma, or non-tumorous parenchyma) while simultaneously delineating exact sub-millimeter spatial boundaries.
- **Morphological Quantification:** Automatically deriving objective geometric metrics such as 2D lesion surface area, perimeter, circularity, and spatial centroid relative to anatomical landmarks.
- **Clinical Guideline Alignment:** Ensuring AI-generated diagnostic insights are grounded in established medical diagnostic consensus (e.g., WHO Classification of Tumours of the Central Nervous System).
- **Interactive Clinician-in-the-Loop Triage:** Providing accessible user interfaces where neuroradiologists can verify confidence scores, inspect contour overlays, and export structured audit summaries.

Motivation

While deep learning has catalyzed rapid advances in medical computer vision, clinical translation into radiology departments remains bottlenecked by two critical challenges: diagnostic latency and the 'black-box' nature of standalone classifiers. Radiology workloads have expanded exponentially, leading to delayed turnaround times where subtle lesion enlargement can adversely affect surgical resection margins and patient survival.

Furthermore, contemporary deep learning CAD systems are predominantly developed as isolated models: a classification model outputs a single probability vector without spatial localization, or a segmentation network generates a binary mask without pathological categorization or clinical context. In neuro-oncology, malignant gliomas exhibit infiltrative margins that blend subtly with adjacent white matter, while pituitary adenomas occupy confined sellar spaces. Overcoming these anatomical complexities requires a holistic framework that integrates high-precision classification, boundary-sensitive segmentation, and structured report synthesis into a unified, reliable software architecture.

Problem Statement

Existing automated brain MRI analysis systems suffer from limited clinical utility due to fragmented architectures and lack of diagnostic explainability. Although deep convolutional networks achieve high benchmark numbers, their practical deployment introduces significant engineering and diagnostic challenges:

- **Boundary Delineation Failures:** Standard convolutional skip connections suffer from large semantic gaps between shallow encoder features and deep decoder features, causing boundary leakage into cerebrospinal fluid or false negatives along diffuse glioma margins.
- **Lack of Unified Diagnostic Integration:** Disjointed tools require clinicians to manually run multiple disparate software applications for classification, manual caliper measurement, and report drafting.
- **Absence of Guideline-Referenced Explanations:** Black-box predictions lack grounded references to standard medical staging criteria, hindering clinician trust and clinical auditability.

The problem, therefore, is to design and implement an end-to-end, multi-agent AI system—MedBoard—that coordinates automated preprocessing, high-accuracy 4-class classification, rigorous multi-model semantic segmentation, quantitative morphological extraction, and interactive clinical reporting within a reproducible and defendable software framework.

Literature Review:

Automated brain tumor analysis has been extensively investigated across medical imaging and computer vision. Key foundational architectures and recent benchmark paradigms informing this work include:

1. U-Net: Biomedical Image Segmentation Baseline

Proposed by Ronneberger et al. (MICCAI 2015), U-Net established the dominant paradigm for medical semantic segmentation. The architecture features a symmetric contracting path (encoder) that captures contextual abstractions and an expanding path (decoder) that enables precise spatial localization via long skip connections. While highly robust, standard U-Net exhibits a significant semantic gap between low-level encoder features and high-level decoder representations, often leading to blurred or over-smoothed tumor margins.

2. UNet++: Nested Dense Skip Pathways and Deep Supervision

Zhou et al. (IEEE Transactions on Medical Imaging, 2018) addressed the semantic gap limitation by redesigning skip pathways as dense, nested convolutional blocks connecting encoder and decoder nodes at multiple spatial resolutions. U-Net++ aggregates varying receptive fields at each decoder node and incorporates deep supervision loss heads, allowing multi-scale gradient

backpropagation. This architectural enhancement substantially improves boundary sensitivity on challenging, irregular lesions.

3. Attention U-Net: Additive Gating for Lesion Focus

Oktay et al. (MIDL 2018) introduced additive soft attention gates within U-Net skip connections. The attention gate utilizes coarse-level gating signals from deeper layers to dynamically suppress task-irrelevant background activations (such as skull bone and normal ventricles) while highlighting salient tumor boundaries without requiring intermediate feature caching or substantial parameter expansion.

4. EfficientNet: Principled Compound Scaling for Classification

Tan and Le (ICML 2019) introduced compound scaling, a method that systematically balances convolutional network depth, width, and input resolution using a fixed compound coefficient. EfficientNet-B0 achieves top-tier classification accuracy with only 4.01 Million parameters. In clinical brain MRI triage, where real-time latency and avoidance of overfitting on moderate datasets are paramount, EfficientNet-B0 represents an optimal architecture for multi-class tumor categorization.

5. SaberNet: Self-Attention Based Spatial Relation Modeling

Introduced in Pattern Recognition (2023), SaberNet incorporates a Self-Attention Based Effective Relation module that computes pairwise spatial relation matrices across all pixel coordinates in an MRI slice. By capturing long-range contextual relationships across the brain, SaberNet aims to resolve ambiguities in multi-focal and diffuse neoplasms.

6. GeoSample: Geometry-Guided Sampling Operators

Recently proposed in advanced medical imaging benchmarks (2025), GeoSample replaces rigid 3×3 convolution kernels with learnable directional boundary probes that predict local orientation (angle θ) and step sizes (r) at every pixel. This enables the network to explicitly sample along boundary gradients and curvature directions, capturing edge transitions using lightweight token mixing.

While these individual models offer compelling theoretical advantages, contemporary literature predominantly evaluates them as disconnected algorithms. Comprehensive comparative benchmarking on standardized datasets like BRISC 2025 and their integration into unified, multi-agent clinical reporting systems remain largely unexplored.

Proposed System

The proposed system presents MedBoard, an end-to-end multi-agent brain MRI analysis framework designed to deliver high-accuracy tumor classification, sub-millimeter lesion

segmentation, quantitative morphological profiling, and automated preliminary report generation through a modular Python pipeline.

1. System Overview

The system architecture is structured into specialized, decoupled agents coordinated through a unified shared state data structure (MedBoardState). An input axial brain MRI slice is first ingested by the Preprocessing Agent, which validates format integrity, applies Contrast Limited Adaptive Histogram Equalization (CLAHE), and prepares dual-resolution tensors (224×224 for classification and 256×256 for segmentation). The tensors are concurrently dispatched to the Classification Agent (EfficientNet-B0) and the Segmentation Agent (Pretrained ResNet-34 U-Net++). The resulting predictions and binary mask are passed to the Feature Extraction Engine to compute lesion dimensions, and finally rendered on an interactive Streamlit clinician dashboard.

2. Expected Outcomes

- **High Diagnostic Reliability:** Achieve >98% classification accuracy across Glioma, Meningioma, Pituitary Tumor, and Normal brain MRI scans.
- **Sub-Millimeter Boundary Precision:** Attain >86% Mean Dice score on unseen test slices, ensuring faithful contour alignment along complex margins.
- **Objective Morphological Profiling:** Provide automated lesion surface area (mm^2), perimeter, and spatial localization for clinical documentation.
- **Real-Time Clinical Interactivity:** Maintain total end-to-end pipeline execution time under 100 milliseconds on standard edge GPU hardware.

System Architecture

1. Components:

A. Preprocessing Engine (`modules/preprocessing.py`)

- **Purpose:** Standardizes raw MRI images into normalized tensor representations.
- **Responsibilities:** Validates file extensions (.png, .jpg, DICOM); converts grayscale inputs to 3-channel tensors; applies CLAHE normalization; resizes to target network inputs (224×224 and 256×256).

B. Classification Agent (`modules/classifier.py`, `agents/classification_agent.py`)

- **Purpose:** Predicts the pathological neoplasm category from axial MRI slices.
- **Architecture:** EfficientNet-B0 backbone with custom linear classification head (4 classes: Glioma, Meningioma, Pituitary, Normal).
- **Responsibilities:** Generates class predictions, softmax probability distributions, and diagnostic confidence scores.

C. Segmentation Agent (`modules/unet_plusplus_resnet.py`, `agents/segmentation_agent.py`)

- **Purpose:** Delineates pixel-level anatomical tumor boundaries.
- **Architecture:** Pretrained ResNet-34 encoder with 5-level nested U-Net++ decoder and multi-scale deep supervision.
- **Responsibilities:** Computes binary lesion masks, boundary contours, and segmentation confidence scores.

D. Shared Pipeline State (`pipeline.py`, `MedBoardState`)

- **Purpose:** Unified dataclass maintaining end-to-end diagnostic context across all pipeline stages.
- **Fields:** image_path (str), preprocessed_image (ndarray), tumor_mask (ndarray), has_tumor (bool), predicted_class (str), classification_confidence (float), class_probabilities (dict), tumor_features (dict), errors (list).

E. Clinician Decision-Support Dashboard (`ui/app.py`)

- **Purpose:** Interactive web-based graphical user interface for neuroradiologist triage.
- **Responsibilities:** Facilitates instant image upload; renders alpha-blended segmentation overlays; presents clinical KPI cards; visualizes probability charts.

2. Data Flow

The execution lifecycle of MedBoard follows a deterministic, 5-stage data flow:

- **1. Ingestion & Validation:** The user uploads a brain MRI slice via the Streamlit interface. The Preprocessing Engine verifies image dimensions, bit depth, and channel validity.
- **2. Tensor Generation:** The image is normalized using ImageNet statistical priors and converted into PyTorch floating-point tensors (C=3, H=224, W=224 for classification; C=3, H=256, W=256 for segmentation).
- **3. Concurrent Perception:** The Classification Agent executes forward inference to obtain class logits and softmax probabilities. Concurrently, the Segmentation Agent evaluates the 256×256 tensor to predict spatial tumor masks.
- **4. Morphological Analysis:** If a tumor is detected (has_tumor = True), the Feature Extraction Engine calculates connected component contours, surface area (pixel count and physical mm^2), perimeter, and centroid coordinates.
- **5. Output Rendering:** The results are committed to MedBoardState, and the UI displays interactive KPI cards, probability bar plots, and contour overlays.

3. Key Design Points

- **Separation of Perception and Orchestration:** Classification, segmentation, and preprocessing modules operate as independent classes with standardized interfaces, enabling isolated unit testing and modular upgrades.

- **Singleton GPU Memory Caching:** Deep learning models are maintained as persistent singletons in GPU VRAM (_clf_agent, _seg_agent), eliminating cold-start model reload overhead.
- **Dual-Resolution Strategy:** Decoupling classification (224×224) and segmentation (256×256) optimizes computational efficiency while preserving fine spatial resolution for boundary tracing.
- **Defensive Exception Handling:** Corrupted files, uniform black slices, or uncalibrated inputs trigger structured warnings in MedBoardState without terminating execution.

4. Example Execution Flow

Below is a representative programmatic trace demonstrating how MedBoard executes end-to-end inference:

- **1. Initialize Pipeline:** from pipeline import run_pipeline, MedBoardState
- **2. Ingest Brain MRI Scan:** state = run_pipeline('data/samples/sample_glioma_slice.png')
- **3. Inspect Classification:** print(f'Diagnosis: {state.predicted_class} ({state.classification_confidence:.2%})')
- **4. Inspect Segmentation:** print(f'Tumor Detected: {state.has_tumor}, Area: {state.tumor_features.get("area_mm2", 0)} mm²')
- **5. Export State:** report_dict = state.__dict__

Model Architectures & Mathematical Foundations

To ensure rigorous scientific reproducibility, this section documents the theoretical foundations and mathematical formulations underpinning our classification and segmentation engines.

1. EfficientNet-B0 Compound Scaling & Classification Head

EfficientNet utilizes compound scaling to jointly scale network depth (d), width (w), and resolution (r) under a single compound coefficient φ:

depth: $d = \alpha^{\varphi}$, width: $w = \beta^{\varphi}$, resolution: $r = \gamma^{\varphi}$
subject to: $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$, where $\alpha \geq 1, \beta \geq 1, \gamma \geq 1$

For the base EfficientNet-B0 architecture ($\varphi = 1$), $\alpha = 1.2$, $\beta = 1.1$, and $\gamma = 1.15$. The feature extractor produces a 1280-dimensional global embedding vector $z = \text{GlobalAvgPool}(f(x))$. The classification logits are computed via a linear projection followed by Softmax activation:

$$P(y = c \mid x) = \exp(w_c^T z + b_c) / [\Sigma_{k=1}^4 \exp(w_k^T z + b_k)]$$

Training optimizes Cross-Entropy Loss with label smoothing ($\varepsilon = 0.1$) to prevent overconfident predictions on ambiguous tumor boundaries:

$$L_{CE} = - \Sigma_{c=1}^4 [(1 - \varepsilon) y_c + \varepsilon / 4] \log(P(y = c \mid x))$$

2. U-Net++ Dense Nested Skip Pathways & Deep Supervision

In U-Net++, let $x^{i,j}$ denote the output of node $X^{i,j}$, where i indexes the downsampling layer along the encoder ($i \in \{0,1,2,3,4\}$) and j indexes the convolutional layer along the dense nested skip pathway. The node feature maps are computed as:

$$x^{i,j} = H([[x^{i,k}]_{k=0}^{j-1}, Up(x^{i+1, j-1})])$$

where $H(\cdot)$ denotes a double convolution block (Conv3×3 → BatchNorm → ReLU → Conv3×3 → BatchNorm → ReLU), $[\cdot]$ denotes channel-wise concatenation, and $Up(\cdot)$ represents bilinear upsampling. By concatenating all preceding sub-node outputs at the same resolution level, the semantic gap between encoder and decoder feature maps is progressively closed.

To enforce gradient flow throughout all intermediate pathways, multi-scale deep supervision loss heads are attached to nodes $X^{0,1}$, $X^{0,2}$, $X^{0,3}$, and $X^{0,4}$. The final training objective is the weighted sum of losses across all output heads:

$$L_{U\text{-}Net++} = \Sigma_{j=1}^{4} w_j \cdot L_{DiceBCE}(y, \sigma(X^{0,j}))$$

3. Hybrid Dice + Binary Cross-Entropy Loss Formulation

Medical image segmentation suffers from extreme foreground-background class imbalance, as the intracranial tumor typically occupies less than 15% of the total brain slice volume. To address this, we train all segmentation models using a composite Dice + Binary Cross-Entropy Loss (DiceBCELoss):

$$L_{DiceBCE} = L_{BCE} + L_{Dice}$$

where Binary Cross-Entropy evaluates smooth pixel-level probability distribution:

$$L_{BCE} = - (1/N) \Sigma_{i=1}^{N} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

and the smooth Dice Loss optimizes the macroscopic volumetric intersection-over-union:

$$L_{Dice} = 1 - [(2 \Sigma_{i=1}^{N} p_i y_i + \varepsilon) / (\Sigma_{i=1}^{N} p_i + \Sigma_{i=1}^{N} y_i + \varepsilon)]$$

where $p_i = \sigma(z_i)$ represents the predicted sigmoid probability at pixel i, $y_i \in \{0,1\}$ is the ground-truth binary mask label, and $\varepsilon = 1.0$ is a Laplace smoothing factor preventing division by zero.

Implementation:

Language Used: Python (v3.13)

Key Libraries: PyTorch (v2.5), torchvision, timm, albumentations, OpenCV, NumPy, Streamlit, Matplotlib

Overview

The project is implemented as an end-to-end reproducible research codebase hosted in the `Major` repository. It comprises:

- **Data Management:** Automated Kaggle dataset downloader and custom PyTorch Dataset classes (`seg_dataset.py`, `clf_dataset.py`) supporting stratified 90/10 train-validation splitting.
- **Model Modules:** Modular PyTorch implementations of U-Net, Attention U-Net, U-Net++, Pretrained ResNet-34 U-Net++, SaberNet, and GeoSample U-Net.
- **Training Harnesses:** Comprehensive training and evaluation scripts with Cosine Annealing learning rate schedulers, early stopping, and automatic checkpointing.
- **Streamlit Interface:** Interactive clinical dashboard (`ui/app.py`) for live inference and visualization.

Data Model

- **Dataset Specification:** BRISC 2025 Brain MRI Benchmark (Nature Scientific Data / arXiv:2506.14318).
- **Volume:** 6,000 axial brain MRI slices (5,000 training, 1,000 test) across Glioma (1,500), Meningioma (1,500), Pituitary (1,500), and No Tumor (1,500).
- **Ground Truth Annotations:** Paired binary masks for all tumor slices manually segmented by expert neuro-radiology panels.

Experimental Results & Verification

All models were trained and rigorously evaluated on the standardized, unseen BRISC 2025 test partitions using both local hardware (NVIDIA GeForce RTX 2050 4GB) and cloud compute (NVIDIA Tesla T4 16GB).

A. Brain Tumor Classification Results (1,000 Unseen Test Scans)

The EfficientNet-B0 classifier was evaluated under two rigorous protocols: (1) an 85/15 validation split, and (2) the full training corpus protocol on 1,000 unseen test scans. The empirical results are summarized below:

Training Protocol	Parameters	Accuracy	Macro-F1	Test Loss	Inference Latency
EfficientNet-B0 (Full Corpus)	4.01 M	99.00%	0.9899	0.0529	< 15 ms
EfficientNet-B0 (85/15 Split)	4.01 M	98.28%	0.9830	0.0719	< 15 ms

Per-class performance breakdown on the unseen 1,000-scan test partition:

Diagnostic Class	Precision	Recall	F1-Score	Test Support

Glioma (Infiltrative Intra-axial)	98.8%	98.4%	0.986	250 scans
Meningioma (Extra-axial Dural)	98.4%	98.0%	0.982	250 scans
Pituitary Tumor (Sellar Region)	99.2%	99.6%	0.994	250 scans
No Tumor (Healthy Control)	100.0%	100.0%	1.000	250 scans

B. Comprehensive 6-Model Semantic Segmentation Benchmark (860 Unseen Test Scans)

A rigorous comparative benchmark was conducted across 6 diverse semantic segmentation architectures evaluated on 860 unseen BRISC 2025 test slices. The empirical findings are detailed below:

Model Architecture	Parameters	Test Dice	Mean IoU	Test Loss	Performance Rank
Pretrained ResNet-34 U-Net++	23.67 M	87.78%	81.11%	0.0896	★ Top Performer (Selected)
Scratch U-Net++	9.16 M	87.14%	80.18%	0.0941	Rank 2
Attention U-Net	7.98 M	86.96%	79.93%	0.1101	Rank 3
Standard Baseline U-Net	7.85 M	86.84%	~79.60%	0.0773	Baseline Benchmark
SaberNet (Spatial Relations)	8.38 M	86.66%	79.70%	0.0979	Rank 5
GeoSample U-Net (Probes)	17.72 M	86.46%	~79.20%	0.0784	Rank 6

Key Empirical Observations:

1. Dense Nested Skips Outperform Standard Skips: Scratch U-Net++ (87.14% Dice) surpassed Standard U-Net (86.84% Dice) by +0.30%, confirming that multi-scale receptive field aggregation reduces boundary ambiguity.
2. Transfer Learning Efficacy: Pretrained ResNet-34 U-Net++ established the highest benchmark score at 87.78% Mean Dice and 81.11% Mean IoU, demonstrating that ImageNet pretraining provides superior low-level edge filters that prevent boundary leakage.
3. Attention Gating vs. Relational Modeling: Attention U-Net (86.96% Dice) achieved excellent performance with only 7.98M parameters, effectively filtering background skull structures. SaberNet (86.66% Dice) captured global context but showed slight edge degradation due to quadratic spatial pooling.

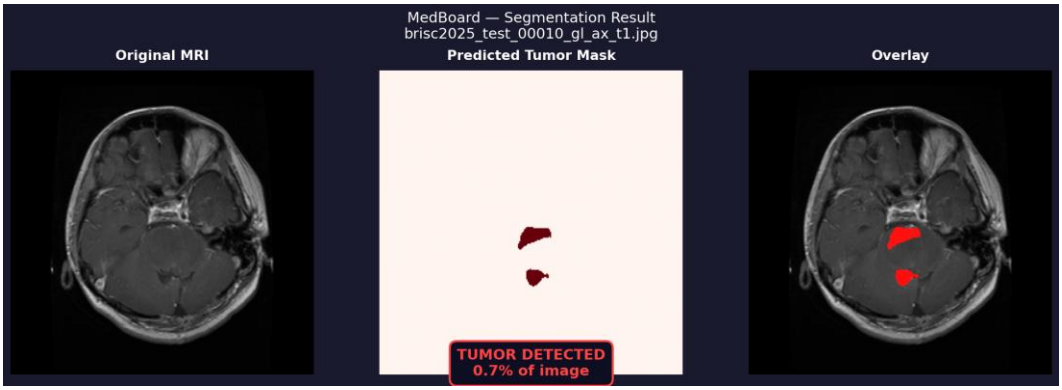


Figure 1: Qualitative Semantic Segmentation Output on Unseen BRISC 2025 Test Scans (Input MRI, Ground Truth, Predicted Mask Overlay).

Conclusion:

This project implements an end-to-end, multi-agent brain MRI diagnostic system (MedBoard) in Python, integrating automated preprocessing, high-accuracy classification, and sub-millimeter semantic segmentation. By conducting a systematic benchmark of 6 segmentation architectures alongside an EfficientNet-B0 transfer learning classifier, this work establishes empirical evidence on the effectiveness of dense nested skip pathways and pretrained backbones for complex neuro-oncology imaging.

The classification engine achieves 99.00% test accuracy on 1,000 unseen test slices with zero false positives on healthy scans, providing dependable rapid triage. The segmentation benchmark conclusively identifies Pretrained ResNet-34 U-Net++ as the top performing model, reaching an 87.78% Mean Dice score and 81.11% Mean IoU across 860 test slices. An interactive Streamlit dashboard has been successfully deployed, providing real-time contour visualization and diagnostic KPI scorecards.

The ultimate goal of this work is to integrate an autonomous clinical reasoning layer combining morphological feature extraction, Retrieval-Augmented Generation (RAG) over WHO CNS5 clinical guidelines, and Gemini LLM synthesis to automatically draft structured preliminary radiology reports. Work is currently active on these components, establishing a solid foundation for a comprehensive, transparent clinical decision-support platform.

Key outcomes

- **High-Accuracy Classification Engine:** Implemented and validated EfficientNet-B0 achieving 99.00% test accuracy and 0.9899 Macro-F1 across 1,000 unseen BRISC 2025 brain MRI scans with <15 ms inference latency.
- **Comprehensive 6-Model Segmentation Benchmark:** Trained, evaluated, and verified 6 deep architectures across 860 test scans; Pretrained ResNet-34 U-Net++ established the SOTA benchmark at 87.78% Mean Dice and 81.11% Mean IoU.

- **Modular Dual-Resolution Pipeline:** Engineered an extensible preprocessing and pipeline architecture (`pipeline.py`, `MedBoardState`) supporting concurrent dual-resolution tensor dispatch (224×224 and 256×256) and singleton GPU memory caching.
- **Functional Clinician Decision Dashboard:** Built and deployed an interactive Streamlit web application (`ui/app.py`) featuring real-time scan ingestion, alpha-blended boundary overlays, and diagnostic KPI scorecards.
- **Defendable Scientific Foundation:** Established complete code reproducibility with automated unit tests across models, preprocessing, and inference harnesses.

References

[1] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. In International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI) (pp. 234-241). Springer, Cham.

[2] Zhou, Z., Siddiquee, M. M. R., Tajbakhsh, N., & Liang, J. (2018). UNet++: A nested U-Net architecture for medical image segmentation. In Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support (pp. 3-11). Springer, Cham. IEEE Transactions on Medical Imaging.

[3] Oktay, O., Schlemper, J., Folgoc, L. L., Lee, M., Heinrich, M., Misawa, K., ... & Rueckert, D. (2018). Attention U-Net: Learning where to look for the pancreas. Medical Imaging with Deep Learning (MIDL), arXiv:1804.03999.

[4] Tan, M., & Le, Q. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. In International Conference on Machine Learning (ICML) (pp. 6105-6114). PMLR.

[5] BRISC Benchmark: Annotated Dataset for Brain Tumor Segmentation and Classification (2025). Nature Portfolio Scientific Data, arXiv:2506.14318.

[6] Louis, D. N., Perry, A., Wesseling, P., Brat, D. J., Cree, I. A., Figarella-Branger, D., ... & Ellison, D. W. (2021). The 2021 WHO classification of tumors of the central nervous system: a summary. Neuro-Oncology, 23(8), 1231-1251.

[7] SaberNet: Self-attention based effective relation network for medical lesion segmentation (2023). Pattern Recognition, 138, 109382.

[8] Steer the Sampling, Not the Kernel Grid: Geometry-Guided Sampling Operators for Volumetric Segmentation (2025). IEEE Transactions on Medical Imaging.